

Lesson 1.2: Dot Products and Orthogonality

Learning Outcomes: At the end of the lesson, the students are able to:

1. compute dot products of two vectors;
2. calculate norms of vectors;
3. establish through proofs the properties of dot product and norm.

Dot Products and Norms

Suppose that we are purchasing items $x_1, x_2, \dots, x_n$ from a supermarket. If the unit price of item x_i is p_i and that we need to purchase q_i units of x_i for $i = 1, 2, \dots, n$, then we may represent these information using the vectors

$$\vec{p} = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{bmatrix} \quad \text{and} \quad \vec{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}$$

where $\vec{p}$ represents the *unit price vector* whose components are the unit price of each item while $\vec{q}$ represents the *quantity vector* whose components are the number of units of each item to be purchased.

If we are interested to calculate the *total price* of the items purchased, we can calculate this using the formula

$$\text{total price} = p_1q_1 + p_2q_2 + \dots + p_nq_n.$$

The total price is what we call as the *dot product* of the vectors $\vec{p}$ and $\vec{q}$.

Definition 12

If $\vec{x}, \vec{y} \in \mathbb{R}^n$, the **dot product** of $\vec{x}$ and $\vec{y}$ is defined as

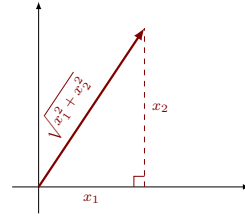
$$\vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i = x_1 y_1 + x_2 y_2 + \dots + x_n y_n,$$

where x_i and y_i are the i th component of $\vec{x}$ and $\vec{y}$, respectively. The **norm** (also

known as **length**) of $\vec{x}$ is defined as

$$\|\vec{x}\| = (\vec{x} \cdot \vec{x})^{1/2} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

We see that $\|\vec{x}\|$ is well-defined since $\vec{x} \cdot \vec{x}$ is a sum of squares of real numbers, hence non-negative. Geometrically, $\|\vec{x}\|$ measures the geometric length of the vector $\vec{x}$.



Example 13. The price of 1 kilo of rice is PhP 50, 1 bar of detergent soap is PhP 30 and 1 can of sardines is PhP 20. How much would 3 kilos of rice, 2 bars of detergent soap and 6 cans of sardines cost?

Solution. We can write the unit price vector and the quantity vector respectively as

$$\vec{p} = \begin{bmatrix} 50 \\ 30 \\ 20 \end{bmatrix}, \text{ and } \vec{q} = \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix}.$$

The total cost is equal to

$$\vec{p} \cdot \vec{q} = (50)(3) + (30)(2) + (20)(6) = \text{PhP } 330. \quad \square$$

Example 14. Given the following vectors in $\mathbb{R}^4$:

$$\vec{u} = \begin{bmatrix} -2 \\ -1 \\ 4 \\ 5 \end{bmatrix}, \vec{v} = \begin{bmatrix} 3 \\ 1 \\ -5 \\ 7 \end{bmatrix}, \text{ and } \vec{w} = \begin{bmatrix} -6 \\ 2 \\ 1 \\ 1 \end{bmatrix}.$$

Calculate each of the following expressions or explain why the expression is undefined:

1. $\|\vec{u} + (\vec{u} \cdot \vec{v})\vec{w}\|$
2. $\|\vec{u}\|^2 \vec{v} + (\vec{v} \cdot \vec{w})\vec{w}$
3. $\|\vec{u} + \vec{v}\| + \vec{w}$

Solutions.

1. To compute $\|\vec{u} + (\vec{u} \cdot \vec{v})\vec{w}\|$:

Step 1: $\vec{u} \cdot \vec{v} = (-2)(3) + (-1)(1) + (4)(-5) + (5)(7) = 8$.

Step 2:

$$\vec{u} + (\vec{u} \cdot \vec{v})\vec{w} = \begin{bmatrix} -2 \\ -1 \\ 4 \\ 5 \end{bmatrix} + 8 \begin{bmatrix} -6 \\ 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -50 \\ 15 \\ 12 \\ 13 \end{bmatrix}.$$

Step 3: $\|\vec{u} + (\vec{u} \cdot \vec{v})\vec{w}\| = \sqrt{(-50)^2 + 15^2 + 12^2 + 13^2} = 7\sqrt{62}$.

2. To compute: $\|\vec{u}\|^2 \vec{v} + (\vec{v} \cdot \vec{w})\vec{w}$

Step 1: $\|\vec{u}\|^2 = \left(\sqrt{(-2)^2 + (-1)^2 + 4^2 + 5^2} \right)^2 = 46$.

Step 2: $\vec{v} \cdot \vec{w} = (3)(-6) + (1)(2) + (-5)(1) + (7)(1) = -14$.

Step 3:

$$\|\vec{u}\|^2 \vec{v} + (\vec{v} \cdot \vec{w})\vec{w} = 46 \begin{bmatrix} 3 \\ 1 \\ -5 \\ 7 \end{bmatrix} + (-14) \begin{bmatrix} -6 \\ 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 222 \\ 18 \\ -244 \\ 308 \end{bmatrix}.$$

3. Observe that $\|\vec{u} + \vec{v}\|$ is a scalar, while $\vec{w}$ is a vector. Since we cannot technically add a scalar to a vector, the expression $\|\vec{u} + \vec{v}\| + \vec{w}$ is *undefined*.



Example 15. Prove that the equation $\|\vec{x} + \vec{y}\| = \|\vec{x}\| + \|\vec{y}\|$ does not hold for all $\vec{x}$ and $\vec{y}$.

Proof. For this, we only need to exhibit a counterexample. Consider the following vectors in $\mathbb{R}^3$:

$$\vec{x} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \text{ and } \vec{y} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}.$$

Note that $\|\vec{x}\| = \|\vec{y}\| = 3$, and that $\vec{x} + \vec{y} = \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}$. Thus,

$$\|\vec{x} + \vec{y}\| = \sqrt{(-1)^2 + 0^2 + 3^2} = \sqrt{10} \neq 3 + 3 = \|\vec{x}\| + \|\vec{y}\|.$$



Theorem 16

For any $\vec{x}, \vec{y}$, and $\vec{z} \in \mathbb{R}^n$,

1. $\vec{x} \cdot \vec{x} \geq 0$ and $\vec{x} \cdot \vec{x} = 0$ if and only if $\vec{x} = \vec{0}$.
2. $\vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{x}$.
3. $\vec{x} \cdot (\vec{y} + \vec{z}) = (\vec{x} \cdot \vec{y}) + (\vec{x} \cdot \vec{z})$.
4. $\vec{x} \cdot (c\vec{y}) = c(\vec{x} \cdot \vec{y})$.

To prove (3), let $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, $\vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$, and $\vec{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}$. Then

$$\vec{x} \cdot (\vec{y} + \vec{z}) = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \cdot \begin{bmatrix} y_1 + z_1 \\ \vdots \\ y_n + z_n \end{bmatrix}$$

Evaluating dot products yields

$$\begin{aligned}
 \vec{x} \cdot (\vec{y} + \vec{z}) &= x_1(y_1 + z_1) + x_2(y_2 + z_2) + \cdots + x_n(y_n + z_n) \\
 &= x_1y_1 + x_1z_1 + x_2y_2 + x_2z_2 + \cdots + x_ny_n + x_nz_n \\
 &= (x_1y_1 + x_2y_2 + \cdots + x_ny_n) + (x_1z_1 + x_2z_2 + \cdots + x_nz_n) \\
 &= \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} + \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \cdot \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \\
 &= (\vec{x} \cdot \vec{y}) + (\vec{x} \cdot \vec{z}). \quad \square
 \end{aligned}$$

We can deduce that $(\vec{x} + \vec{y}) \cdot \vec{z} = (\vec{x} \cdot \vec{z}) + (\vec{y} \cdot \vec{z})$ and $(c\vec{x}) \cdot \vec{y} = c(\vec{x} \cdot \vec{y})$ using properties (2) - (4). The proofs of other properties are left as exercises to the student. We next state a number of properties of the vector norm.

Theorem 17

For any $\vec{x} \in \mathbb{R}^n$ and for any $c \in \mathbb{R}$,

1. $\|\vec{x}\|^2 = \vec{x} \cdot \vec{x}$.
2. $\|\vec{x}\| \geq 0$ and $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$.
3. $\|c\vec{x}\| = |c| \|\vec{x}\|$.

Proof. Again, we shall only prove (3) and leave the others as exercises.

Let $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and let $c \in \mathbb{R}$. Then $c\vec{x} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}$. Thus,

$$\begin{aligned}
 \|c\vec{x}\| &= \sqrt{(cx_1)^2 + (cx_2)^2 + \cdots + (cx_n)^2} = \sqrt{c^2x_1^2 + c^2x_2^2 + \cdots + c^2x_n^2} \\
 &= \sqrt{c^2(x_1^2 + x_2^2 + \cdots + x_n^2)} = \sqrt{c^2} \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \\
 &= |c| \|\vec{x}\|.
 \end{aligned}$$



Example 18. Let $\vec{x}$ and $\vec{y}$ be vectors. Prove that $(\vec{x} + \vec{y}) \cdot (\vec{x} - \vec{y}) = 0$ if and only if $\|\vec{x}\| = \|\vec{y}\|$.

Proof. Using the properties of dot product and norm, note that

$$\begin{aligned} (\vec{x} + \vec{y}) \cdot (\vec{x} - \vec{y}) &= \vec{x} \cdot (\vec{x} - \vec{y}) + \vec{y} \cdot (\vec{x} - \vec{y}) \\ &= \vec{x} \cdot \vec{x} - \vec{x} \cdot \vec{y} + \vec{y} \cdot \vec{x} - \vec{y} \cdot \vec{y} \\ &= \|\vec{x}\|^2 - \vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{y} - \|\vec{y}\|^2 = \|\vec{x}\|^2 - \|\vec{y}\|^2. \end{aligned}$$

Thus, $(\vec{x} + \vec{y}) \cdot (\vec{x} - \vec{y}) = 0$ if and only if $\|\vec{x}\|^2 - \|\vec{y}\|^2 = 0$, or equivalently, $\|\vec{x}\|^2 = \|\vec{y}\|^2$. Since norms are non-negative,

$$(\vec{x} + \vec{y}) \cdot (\vec{x} - \vec{y}) = 0 \text{ if and only if } \|\vec{x}\| = \|\vec{y}\|.$$



A vector $\vec{u}$ is called a **unit vector** if $\|\vec{u}\| = 1$. As an example, the vector

$$\vec{u} = \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix}$$

in $\mathbb{R}^3$ is a unit vector since

$$\|\vec{u}\| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2} = 1.$$

Two nonzero vectors are said to be in the **same direction** if one is a positive scalar multiple of the other. Given a nonzero vector $\vec{x}$, the next result gives a formula for the unit vector in the same direction as $\vec{x}$ called as the **normalized** multiple of $\vec{x}$.

Theorem 19

Let $\vec{x}$ be a nonzero vector. The vector

$$\vec{u} = \frac{\vec{x}}{\|\vec{x}\|} := \frac{1}{\|\vec{x}\|} \vec{x}$$

is a unit vector in the same direction as $\vec{x}$.

Proof. If $\vec{x} \neq \vec{0}$, then $\|\vec{x}\| > 0$ so $1/\|\vec{x}\| > 0$. Thus, $\vec{u} = (1/\|\vec{x}\|)\vec{x}$ is in the same direction as $\vec{x}$. To show that $\vec{u}$ is a unit vector, one checks that

$$\|\vec{u}\| = \left\| \frac{1}{\|\vec{x}\|} \vec{x} \right\| = \frac{1}{\|\vec{x}\|} \|\vec{x}\| = 1. \quad \square$$

Example 20. Let $\vec{x} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$. Find the vector of length 5 which is in the same direction as $\vec{x}$.

Solution. Note that $\|\vec{x}\| = \sqrt{2^2 + (-1)^2 + 2^2} = 3$. Hence,

$$\frac{\vec{x}}{\|\vec{x}\|} = \frac{1}{3} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2/3 \\ -1/3 \\ 2/3 \end{bmatrix}$$

is a unit vector in the direction of $\vec{x}$. Multiplying this resulting vector by 5 gives

$$5 \begin{bmatrix} 2/3 \\ -1/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 10/3 \\ -5/3 \\ 10/3 \end{bmatrix},$$

which is a vector of length 5 in the same direction as $\vec{x}$. \(\square\)

We next prove a result which plays an important role in linear algebra and on a number of areas in mathematics. Briefly, it tells us that two vectors of *short* lengths cannot have a *large* dot product.

Theorem 21: Cauchy-Schwarz Inequality

For any $\vec{x}, \vec{y} \in \mathbb{R}^n$, $|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \|\vec{y}\|$.

Proof. If either $\vec{x} = \vec{0}$ or $\vec{y} = \vec{0}$, then theorem holds trivially. Assume that neither $\vec{x}$ nor $\vec{y}$ is $\vec{0}$. Consider the function $g(t) := \|\vec{x}t - \vec{y}\|^2$ for $t \in \mathbb{R}$. Note that $g(t) \geq 0$ for all $t \in \mathbb{R}$.

Notice that we may write

$$\begin{aligned} g(t) &= (t\vec{x} - \vec{y}) \cdot (t\vec{x} - \vec{y}) \\ &= \|\vec{x}\|^2 t^2 - 2(\vec{x} \cdot \vec{y})t + \|\vec{y}\|^2 \\ &= at^2 + bt + c, \end{aligned}$$

where $a = \|\vec{x}\|^2$, $b = -2(\vec{x} \cdot \vec{y})$, and $c = \|\vec{y}\|^2$. Since $a > 0$, the quadratic function $g(t)$ has at most one real root implying that its discriminant $b^2 - 4ac \leq 0$. This inequality implies

$$4(\vec{x} \cdot \vec{y})^2 - 4\|\vec{x}\|^2\|\vec{y}\|^2 \leq 0.$$

Therefore, $|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\|\|\vec{y}\|$ as required. ▮

Example 22. If $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n \in \mathbb{R}$, prove that

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right).$$

Proof. If we let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$, the Cauchy-Schwarz inequality implies that

$$|x_1 y_1 + x_2 y_2 + \dots + x_n y_n| \leq \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} \sqrt{y_1^2 + y_2^2 + \dots + y_n^2}.$$

Squaring both sides yields

$$(x_1 y_1 + x_2 y_2 + \dots + x_n y_n)^2 \leq (x_1^2 + x_2^2 + \dots + x_n^2)(y_1^2 + y_2^2 + \dots + y_n^2),$$

which is what we want. ▮

The following follows directly from the Cauchy-Schwarz inequality.

Corollary 23: Triangle Inequality

For any vectors $\vec{x}, \vec{y}$ in $\mathbb{R}^n$, $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$.

Proof. This is a direct application of the Cauchy-Schwarz inequality.

$$\begin{aligned}\|\vec{x} + \vec{y}\|^2 &= \|\vec{x}\|^2 + 2(\vec{x} \cdot \vec{y}) + \|\vec{y}\|^2 \\ &\leq \|\vec{x}\|^2 + 2\|\vec{x}\|\|\vec{y}\| + \|\vec{y}\|^2 \\ &= (\|\vec{x}\| + \|\vec{y}\|)^2.\end{aligned}$$

**Angle Between Vectors and Orthogonality**

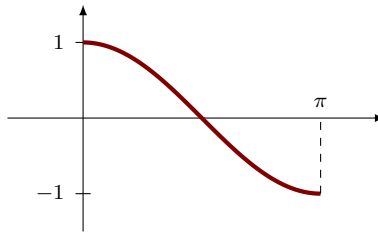
Let $\vec{x}$ and $\vec{y}$ be nonzero vectors. The Cauchy-Schwarz inequality may be written alternatively as,

$$-(\|\vec{x}\|\|\vec{y}\|) \leq \vec{x} \cdot \vec{y} \leq \|\vec{x}\|\|\vec{y}\|.$$

Since $\|\vec{x}\| > 0$ and $\|\vec{y}\| > 0$, it follows that

$$-1 \leq \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|} \leq 1.$$

Observe that the cosine function is one-to-one or injective on the interval $[0, \pi]$ as its value decreases continuously from 1 to -1 .



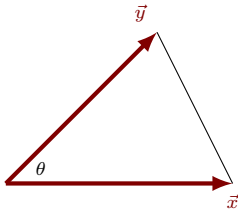
Hence, there exists a unique $\theta \in [0, \pi]$ such that $\cos \theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|}$.

Definition 24: Angle Between Vectors

If $\vec{x}$ and $\vec{y}$ are nonzero vectors in $\mathbb{R}^n$, the **angle** between $\vec{x}$ and $\vec{y}$ is the unique $\theta \in [0, \pi]$ such that

$$\cos \theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|}.$$

To make a sense of this definition, we consider the case when $n = 2$. We think of θ as the geometric angle between the arrow representations of $\vec{x}$ and $\vec{y}$ as shown.



We see that vectors $\vec{x}$ and $\vec{y}$ define a triangle whose sides are $\|\vec{x}\|$, $\|\vec{y}\|$, and $\|\vec{x} - \vec{y}\|$. By the *law of cosines*, we see that

$$\|\vec{x} - \vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2 - 2\|\vec{x}\| \|\vec{y}\| \cos \theta.$$

Solving for $\cos \theta$ in this equation gives

$$\cos \theta = \frac{\|\vec{x}\|^2 + \|\vec{y}\|^2 - \|\vec{x} - \vec{y}\|^2}{2\|\vec{x}\| \|\vec{y}\|}.$$

Taking note that $\|\vec{x} - \vec{y}\|^2 = \|\vec{x}\|^2 - 2(\vec{x} \cdot \vec{y}) + \|\vec{y}\|^2$ and plugging this in the above equation gives

$$\cos \theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|}.$$

Example 25. Let θ be the angle between the vectors

$$\vec{x} = \begin{bmatrix} 0 \\ -2 \\ -1 \\ 1 \end{bmatrix} \quad \text{and} \quad \vec{y} = \begin{bmatrix} -3 \\ 2 \\ 4 \\ 4 \end{bmatrix}.$$

Find $\cos \theta$.

Solution. Note that

$$\bullet \quad \vec{x} \cdot \vec{y} = (0)(-3) + (-2)(2) + (-1)(4) + (1)(4) = -4;$$

- $\|\vec{x}\| = \sqrt{0^2 + (-2)^2 + (-1)^2 + 1^2} = \sqrt{6}$; and
- $\|\vec{y}\| = \sqrt{(-3)^2 + 2^2 + 4^2 + 4^2} = 3\sqrt{5}$.

If θ is the angle between $\vec{x}$ and $\vec{y}$, then

$$\cos \theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{-4}{\sqrt{6} \cdot 3\sqrt{5}} = -\frac{4}{3\sqrt{30}}.$$



Example 26. Find the angle between the diagonal of a cube and one of its edges.

Solution. Without loss of generality, consider a cube of side length a in $\mathbb{R}^3$ whose sides are parallel to the axes so that $(0, 0, 0)$ and (a, a, a) are vertices of a diagonal, where $a > 0$. Notice that the angle θ required in the problem is the angle between the vectors

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } \vec{d} = \begin{bmatrix} a \\ a \\ a \end{bmatrix}.$$

From the angle formula,

$$\cos \theta = \frac{\vec{d} \cdot \vec{e}_1}{\|\vec{d}\| \|\vec{e}_1\|} = \frac{a}{\sqrt{a^2 + a^2 + a^2} \sqrt{1 + 0 + 0}} = \frac{a}{a\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

It now follows that $\theta = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right) \approx 54.74^\circ$.



Looking at the cosine formula for angles

$$\cos \theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|},$$

the denominator of the fraction is always positive. Hence, the sign of the dot product $\vec{x} \cdot \vec{y}$ determines the sign of $\cos \theta$. We get the following as remarks.

Let $\theta \in [0, \pi]$ be the angle between the nonzero vectors $\vec{x}$ and $\vec{y}$ in $\mathbb{R}^n$. Then

- $\theta = \pi/2$ (right) if and only if $\vec{x} \cdot \vec{y} = 0$.
- $\theta < \pi/2$ (acute) if and only if $\vec{x} \cdot \vec{y} > 0$.
- $\theta > \pi/2$ (obtuse) if and only if $\vec{x} \cdot \vec{y} < 0$.

With these observations, we have the following definition.

Definition 27

Two vectors $\vec{x}$ and $\vec{y}$ are **orthogonal** if $\vec{x} \cdot \vec{y} = 0$. A nonempty set of vectors is **orthogonal** if every pair of vectors in the set are orthogonal. An orthogonal set of vectors is said to be **orthonormal** if every vector from the set is a unit vector.

Example 28. The set $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$, where

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} -2 \\ 1 \\ -4 \\ 2 \end{bmatrix}, \text{ and } \vec{v}_3 = \begin{bmatrix} -4 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

is an orthogonal set of vectors in $\mathbb{R}^4$ since

$$\vec{v}_1 \cdot \vec{v}_2 = 0, \vec{v}_1 \cdot \vec{v}_3 = 0, \text{ and } \vec{v}_2 \cdot \vec{v}_3 = 0$$

Notice also that $\|\vec{v}_1\| = \|\vec{v}_2\| = \|\vec{v}_3\| = 5$, so S is not orthonormal. However, if we let

$$\vec{w}_1 = \frac{1}{5} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \end{bmatrix}, \vec{w}_2 = \frac{1}{5} \begin{bmatrix} -2 \\ 1 \\ -4 \\ 2 \end{bmatrix}, \text{ and } \vec{w}_3 = \frac{1}{5} \begin{bmatrix} -4 \\ 2 \\ 2 \\ 1 \end{bmatrix},$$

then the set $T = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ is orthonormal.

Example 29. The unit vectors $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ form an orthonormal set in $\mathbb{R}^3$. More generally, the standard unit vectors in $\mathbb{R}^n$, namely,

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \vec{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

form an orthonormal set in $\mathbb{R}^n$.

Remark. The set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is orthonormal if and only if $\vec{v}_i \cdot \vec{v}_j = \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j; \\ 0, & \text{if } i \neq j. \end{cases}$$

In linear algebra, δ_{ij} is called as the **Kronecker delta**.

The following is an n -dimensional version of the *Pythagorean theorem* which states that the sum of the squares of the norms of a pair of orthogonal vectors is equal to the square of the norm of their sum.

Theorem 30: Pythagorean Theorem

Let $\vec{x}$ and $\vec{y}$ be vectors in $\mathbb{R}^n$. Then

$$\vec{x} \cdot \vec{y} = 0 \text{ if and only if } \|\vec{x} + \vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2.$$

Proof. The theorem simply follows from the fact that

$$\|\vec{x} + \vec{y}\|^2 = (\vec{x} + \vec{y}) \cdot (\vec{x} + \vec{y}) = \|\vec{x}\|^2 + 2(\vec{x} \cdot \vec{y}) + \|\vec{y}\|^2. \quad \square$$